National University of Computer and Emerging Sciences

Lab Exercise 10
AL2002-Artifical Intelligence Lab

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Exercise

1. Implement a single-layer perceptron to model the AND logic gate.

x_1	x_2	y_{AND}
0	0	0
0	1	0
1	0	0
1	1	1

   Do not use built in functions.

```
# Step function (Activation function)
def step_function(z)

# Perceptron Learning Algorithm
def train_perceptron(X, y, epochs=10, learning_rate=0.1)

# Predict function
def predict(X, weights, bias)

#Plot decision boundary
def plot_decision_boundary(X, y, weights, bias, title)
```

You can get help here:
https://medium.com/analytics-vidhya/implementing-perceptron-learning-algorithm-to-solve-and-in-python-903516300b2f

2. Implementation of a Single-Layer Perceptron using TensorFlow
   - Loads the given dataset.

```
from sklearn.datasets import load_breast_cancer
X, y = load_breast_cancer(return_X_y=True)
```

   - Follow the steps provided in the attached manual to train a single layer perceptron on this dataset.
   - Evaluate your model using appropriate evaluation metrics.
   - Comment on your model performance.